

Rechargeable batteries will be an important part of a fossil fuel-free future, because they can store electricity from renewable sources.

Electrifying the future

Akira discovered that using carbon in a lithium-ion battery made it safer than earlier versions. He applied for a patent for his battery in 1985. A patent is a special document that describes what is new about an invention and gives the owner the right to be the only one to sell it for a certain time. The first lithium-ion batteries were sold in 1991—around 10 years after development began!

Lithium-ion batteries have continued to get thinner and lighter. They are now found in laptops, phones, tablets, and games consoles. Larger versions have been key in developing electric vehicles. Today, almost 65 percent of the lithium produced each year is used to make batteries. There are still technological and environmental problems to solve, but Akira's work on this rechargeable energy source has paved the way toward a future free from fossil fuels, such as oil.



In 2019, Akira shared the Nobel Prize in Chemistry for his work.